Problem set 3

Solution Key

Econ 5030

The question text is reproduced in the grey boxes. Everything after a box is the answer.

1 Basic Cournot duopoly

Let begin by considering a Cournot duopoly. Each firm i has the cost function $C(q_i) = cq_i^2/2$, where $c > 0$. Market inverse $P(Q) = 200 - Q$.

1.a *Write the profit of the arbitrary firm i.*

With $Q = q_i + q_j$,

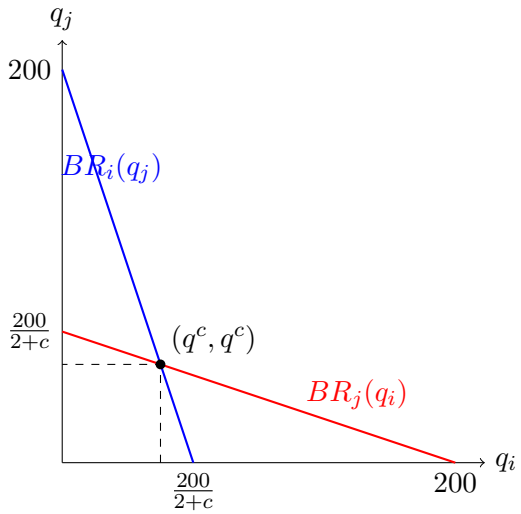
$$\pi_i(q_i, q_j) = (200 - q_i - q_j)\, q_i - \frac{c}{2} q_i^2.$$

1.b *Find the best response function for firm i and graph it on the $q_i \times q_j$ plane.*

π_i is strictly concave in q_i ($\partial^2 \pi_i / \partial q_i^2 = -(2 + c) < 0$), so the first-order condition is sufficient:

$$\frac{\partial \pi_i}{\partial q_i} = 200 - 2q_i - q_j - cq_i = 0 \quad \Longrightarrow \quad BR_i(q_j) = \max\left\{\frac{200 - q_j}{2 + c},\, 0\right\}.$$

It is a downward-sloping line with q_i-intercept $200/(2 + c)$ (the monopoly output) and q_j-intercept 200 (the competitive output, where price would hit zero). Quantities are strategic substitutes: the slope is $-1/(2 + c)$.



***1.c** Find the Cournot equilibrium quantity q^c and market price $p^c = P(2q^c)$.*

The game is symmetric, so set $q_i = q_j = q^c$ in the best response: $(2+c)q^c = 200 - q^c$, so

$$q^c = \frac{200}{3+c}, \qquad Q^c = \frac{400}{3+c}, \qquad p^c = 200 - \frac{400}{3+c} = \frac{200(1+c)}{3+c}.$$

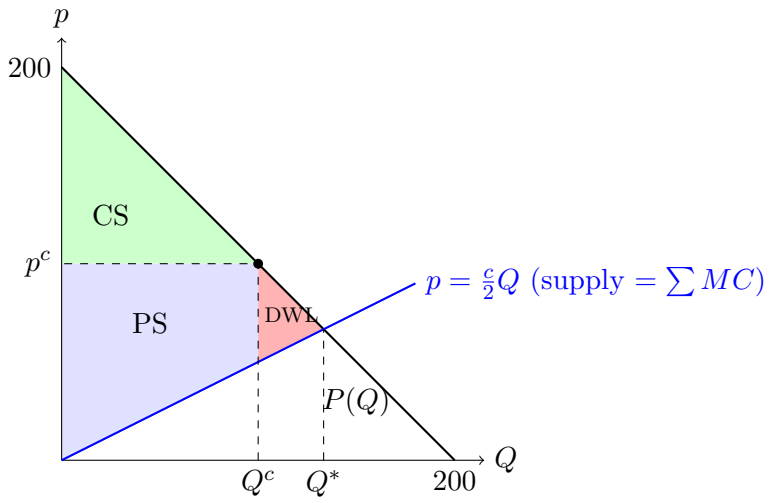
(The equilibrium is unique: the two best-response lines have slopes $-1/(2+c)$ and $-(2+c)$, so they cross once.)

***1.d** Graph the market outcome on the $p \times Q$ plane.*

Each firm's marginal cost is $MC_i = cq_i$, so firm supply is $q_i = p/c$ and industry supply (the sum of marginal cost curves) is $Q = 2p/c$, i.e. $p = cQ/2$. The efficient point is where demand meets this supply curve:

$$200 - Q = \tfrac{c}{2}Q \implies Q^* = \frac{400}{2+c}, \qquad p^* = \frac{200c}{2+c}.$$

Cournot output $Q^c < Q^*$ and price $p^c > p^*$.



(The picture uses $c = 1$ for scale. CS is the triangle above p^c, PS is the area between p^c and the supply curve up to Q^c, and the deadweight loss is the triangle between demand and supply from Q^c to Q^*.)

***1.e** Find an expression for the each firm's profits, the Consumer Surplus (CS), Total Surplus (TS) and Dead weight Loss (DWL).*

Profit. $\pi^c = p^c q^c - \frac{c}{2}(q^c)^2 = q^c\left(p^c - \frac{c}{2}q^c\right) = \frac{200}{3+c} \cdot \frac{200(1+c)-100c}{3+c}$, so

$$\pi^c = \frac{20000(2+c)}{(3+c)^2}.$$

Consumer surplus. Demand has slope -1, so $CS = \frac{1}{2}(200 - p^c)Q^c = \frac{1}{2}(Q^c)^2$:

$$CS = \frac{80000}{(3+c)^2}.$$

Total surplus. $TS = CS + 2\pi^c = \dfrac{80000 + 40000(2+c)}{(3+c)^2}$, so

$$TS = \frac{40000(4+c)}{(3+c)^2}.$$

Deadweight loss. The maximum total surplus is the triangle between demand and supply from 0 to Q^*: $TS^* = \frac{1}{2} \cdot 200 \cdot Q^* = \dfrac{40000}{2+c}$. Then

$$DWL = TS^* - TS = 40000\,\frac{(3+c)^2 - (4+c)(2+c)}{(2+c)(3+c)^2} = \frac{40000}{(2+c)(3+c)^2}.$$

Check ($c = 2$, i.e. $C = q^2$): $q^c = 40$, $p^c = 120$, $\pi^c = 3200$, $CS = 3200$, $TS = 9600$, $TS^* = 10000$, $DWL = 400$.

2 Basic Cournot oligopoly

Let now consider a Cournot oligopoly with n firms. Each firm i has the cost function $C(q_i) = cq_i^2/2$, *where* $c > 0$. *Market inverse* $P(Q) = 200 - Q$.

2.a *Write the profit of the arbitrary firm i.*

With $Q_{-i} = \sum_{j \neq i} q_j$,

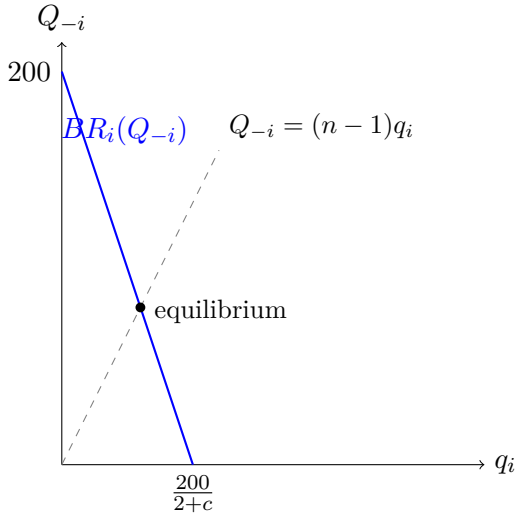
$$\pi_i(q_i, Q_{-i}) = (200 - q_i - Q_{-i})\, q_i - \frac{c}{2} q_i^2.$$

2.b *Find the best response function for firm i and graph it on the* $q_i \times Q_{-i}$ *plane.*

Same algebra as 1.b with q_j replaced by Q_{-i} (still strictly concave):

$$200 - 2q_i - Q_{-i} - cq_i = 0 \quad \Longrightarrow \quad BR_i(Q_{-i}) = \max\left\{\frac{200 - Q_{-i}}{2+c},\, 0\right\}.$$

Firm i cares only about the total output of its rivals, not how it is split.



In a symmetric equilibrium the best-response line meets the ray $Q_{-i} = (n-1)q_i$. (Drawn for $c = 1$, $n = 3$.)

2.c *Find the Cournot equilibrium quantity q^c and market price $p^c = P(nq^c)$.*

In the symmetric equilibrium $Q_{-i} = (n-1)q^c$, so $(2+c)q^c = 200 - (n-1)q^c$:

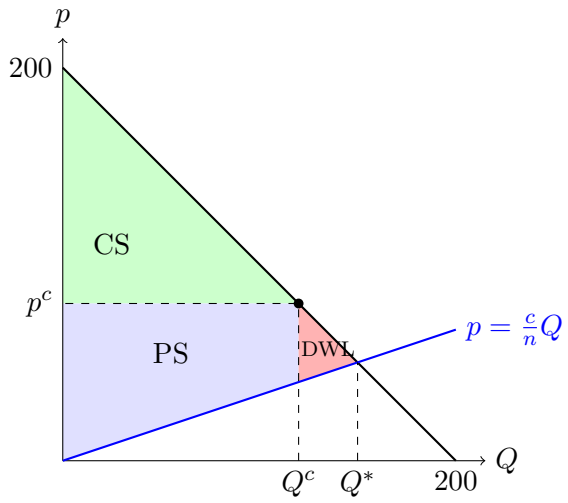
$$q^c = \frac{200}{n+1+c}, \qquad Q^c = \frac{200n}{n+1+c}, \qquad p^c = 200 - Q^c = \frac{200(1+c)}{n+1+c}.$$

With $n = 2$ this is 1.c. As $n \to \infty$: $q^c \to 0$, $Q^c \to 200$ and $p^c \to 0$.

2.d *Graph the market outcome on the $p \times Q$ plane.*

Industry supply is now $Q = np/c$, i.e. $p = \frac{c}{n}Q$, which gets flatter as n grows. The efficient point is

$$200 - Q = \tfrac{c}{n}Q \implies Q^* = \frac{200n}{n+c}, \qquad p^* = \frac{200c}{n+c}.$$



Same picture as 1.d: CS above p^c, PS between p^c and the supply curve, DWL the triangle from Q^c to Q^*. As n rises, Q^c moves toward Q^* and the triangle shrinks. (Drawn for $c = 1$, $n = 3$.)

2.e *Find an expression for the each firm's profits, the Consumer Surplus (CS), Total Surplus (TS) and Dead weight Loss (DWL).*

Profit. $\pi^c = q^c\big(p^c - \frac{c}{2}q^c\big) = \dfrac{200}{n+1+c} \cdot \dfrac{200(1+c) - 100c}{n+1+c}$, so

$$\pi^c = \frac{20000(2+c)}{(n+1+c)^2}.$$

Consumer surplus. $CS = \frac{1}{2}(Q^c)^2 = \dfrac{20000\,n^2}{(n+1+c)^2}$.

Total surplus. $TS = CS + n\pi^c = \dfrac{20000\,n^2 + 20000\,n(2+c)}{(n+1+c)^2} = \dfrac{20000\,n(n+2+c)}{(n+1+c)^2}$.

Deadweight loss. $TS^* = \frac{1}{2} \cdot 200 \cdot Q^* = \dfrac{20000\,n}{n+c}$. Let $m = n + c$. Then $(m+1)^2 - m(m+2) = 1$, so

$$DWL = 20000\,n\Big[\frac{1}{n+c} - \frac{n+2+c}{(n+1+c)^2}\Big] = \frac{20000\,n}{(n+c)(n+1+c)^2}.$$

Checks. With $n = 2$, every formula reduces to 1.e. As $n \to \infty$, $DWL \sim 20000/n^2 \to 0$ and $\pi^c \to 0$: more competition moves the market toward the efficient outcome. Example ($c = 1$, $n = 4$): $q^c = 33.3$, $p^c = 66.7$, $\pi^c = 1666.7$, $DWL = 444.4$.

3 Cournot with product differentiation

Take the inverse demand system:

$$P_1(q_1, q_2) = 18 - q_1 - \gamma q_2, \qquad P_2(q_2, q_1) = 18 - q_2 - \gamma q_1.$$

Assume both firms have NO COST of production. Now let us look at the Cournot solution for general γ between zero an one.

The parts are printed as 2.a–2.d in the problem set. They are labelled 3.a–3.d here.

3.a *Write out each firm's profit function and label them $\Pi_1(\gamma)$ and $\Pi_2(\gamma)$.*

$$\Pi_1(\gamma) = (18 - q_1 - \gamma q_2)\, q_1, \qquad \Pi_2(\gamma) = (18 - q_2 - \gamma q_1)\, q_2.$$

3.b *Find each firm's best response function by solving $\dfrac{\partial \Pi_1(\gamma)}{\partial q_1} = 0$ for $q_1^\gamma(q_2)$ and $\dfrac{\partial \Pi_2(\gamma_2, \gamma_1)}{\partial q_2} = 0$ for $q_2^\gamma(q_1)$.*

Each profit is strictly concave in own output (second derivative -2).

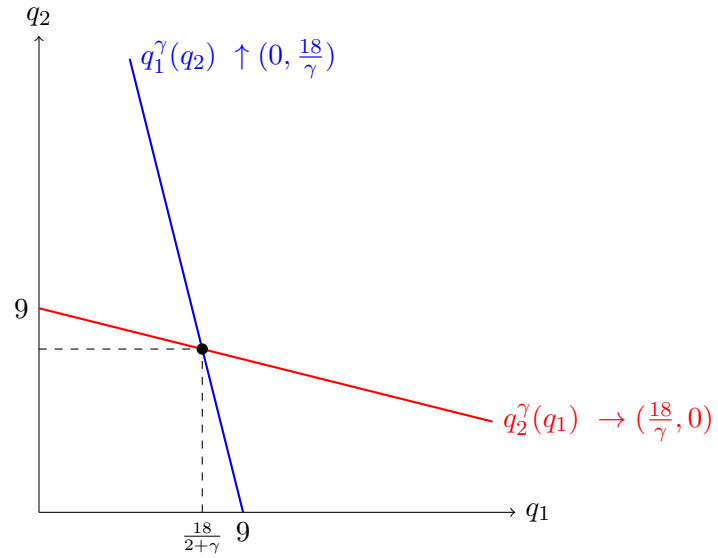
$$\frac{\partial \Pi_1}{\partial q_1} = 18 - 2q_1 - \gamma q_2 = 0 \implies q_1^\gamma(q_2) = \frac{18 - \gamma q_2}{2},$$

$$\frac{\partial \Pi_2}{\partial q_2} = 18 - 2q_2 - \gamma q_1 = 0 \implies q_2^\gamma(q_1) = \frac{18 - \gamma q_1}{2}.$$

The parameter γ measures how much the rival's output lowers my price, i.e. how close substitutes the goods are. $\gamma = 0$: independent goods (two monopolies). $\gamma = 1$: perfect substitutes (homogeneous Cournot).

3.c *Graph the best response functions of the firms' on $q_1 \times q_2$ space.*

$q_1^\gamma(q_2)$ runs from $(9, 0)$ to $(0, 18/\gamma)$; $q_2^\gamma(q_1)$ runs from $(0, 9)$ to $(18/\gamma, 0)$. A larger γ makes each line steeper relative to its own axis and pulls the crossing toward the origin.



(Drawn for $\gamma = \frac{1}{2}$, where the $18/\gamma = 36$ intercepts are off the picture.)

3.d *Find the equilibrium quantities and prices and label them $q_1(\gamma)$, $p_1(\gamma)$ and $q_2(\gamma)$, $p_2(\gamma)$.*

By symmetry $q_1 = q_2 = q$: $2q = 18 - \gamma q$, so

$$q_1(\gamma) = q_2(\gamma) = \frac{18}{2+\gamma}, \qquad p_1(\gamma) = p_2(\gamma) = 18 - (1+\gamma)\frac{18}{2+\gamma} = \frac{18}{2+\gamma}, \qquad \Pi_i = \frac{324}{(2+\gamma)^2}.$$

(Direct check without imposing symmetry: subtracting the two first-order conditions gives $(2-\gamma)(q_1 - q_2) = 0$, so $q_1 = q_2$ since $\gamma < 2$.)

Price, quantity and profit all fall as γ rises. At $\gamma = 0$: $q = p = 9$, $\Pi = 81$ (each firm is a monopolist). At $\gamma = 1$: $q = p = 6$, $\Pi = 36$ (homogeneous Cournot). Product differentiation softens competition.

4 Price competition

Suppose that thre are 2 firms in a Bertrand oligopoly with homogenous products. Each firm i has th cost of production $c(q_i) = q_i^2/2$. The industry demand function is the rectangle

$$D(p) = \begin{cases} 0 & \text{if } p > 10 \\ 10 & \text{if } p \leq 10 \end{cases}$$

The if firms price the same then demand is split evenly. Each firm i has th cost of production $c(q_i) = q_i^2/2$. Based on this cost structure a firm will never supply more than their supply function $s(p_i) = p_i$. using this fact we can construct residual demand and firm profit. Firm profit is the piecewise function for prices in the interval $[0, 10]$ (I have done all the work for you here):

$$\pi_i(p_i, p_j) = \begin{cases} p_i \min\{p_i, 10 - p_j\} - \left(\min\{p_i, 10 - p_j\}\right)^2/2 & p_i > p_j \\ p_i 5 - 12.5 & p_i = p_j > 5 \\ p_i^2/2 & p_i = p_j \leq 5 \\ p_i^2/2 & p_i < p_j \end{cases}$$

Two facts used throughout:

(A) *Low price:* $p_i < p_j$ gives $p_i^2/2$, strictly increasing in p_i.

(B) *High price:* for $p_i > p_j$, profit is increasing in p_i on both pieces of the min and continuous where they meet, so the best high price is $p_i = 10$. There the firm sells the residual $10 - p_j$:

$$\pi_i(10, p_j) = 10(10 - p_j) - \tfrac{1}{2}(10 - p_j)^2 = \frac{(10 - p_j)(10 + p_j)}{2} = \frac{100 - p_j^2}{2}.$$

1. *Because for most prices p_j the best response is not well-defined, instead find the argument of the supremum of the profit responding to the other firm's price. Formally find*

$$\phi(p_j) = \arg \sup_{p_i \in [0,10]} \pi_i(p_i, p_j)$$

Compare the three options against p_j:

- Undercut ($p_i < p_j$): by (A) the supremum is $p_j^2/2$, approached as $p_i \uparrow p_j$ but not attained.
- Match ($p_i = p_j$): $p_j^2/2$ if $p_j \le 5$; $5p_j - 12.5$ if $p_j > 5$. Since $p_j^2/2 - (5p_j - 12.5) = \frac{1}{2}(p_j - 5)^2 > 0$ for $p_j > 5$, matching is never better than undercutting.
- Go high ($p_i = 10$): $(100 - p_j^2)/2$ by (B), available when $p_j < 10$.

Going high beats undercutting iff $100 - p_j^2 > p_j^2$, i.e. $p_j < \sqrt{50} = 5\sqrt{2} \approx 7.07$. Therefore

$$\phi(p_j) = \begin{cases} \{10\} & 0 \le p_j \le 5\sqrt{2} \quad (\text{at } p_j = 5\sqrt{2} \text{ both give } 25) \\ p_j^- \text{ (just below } p_j) & 5\sqrt{2} < p_j \le 10 \end{cases}$$

where p_j^- means the supremum $p_j^2/2$ is approached by $p_j - \varepsilon$, $\varepsilon \downarrow 0$, and is not attained. This is why a best response does not exist for $p_j > 5\sqrt{2}$.

2. *Make a formal argument that symmetric prices cannot be a pure strategy Nash equilibrium.*

Suppose $p_1 = p_2 = p$.

- If $p \le 5$: each earns $p^2/2 \le 12.5$. Deviating to 10 gives $(100 - p^2)/2 \ge 37.5 > p^2/2$. Profitable.
- If $5 < p \le 10$: each earns $5p - 12.5$. Undercutting to $p - \varepsilon$ gives $(p - \varepsilon)^2/2 \to p^2/2$, and $p^2/2 - (5p - 12.5) = \frac{1}{2}(p - 5)^2 > 0$, so a small enough ε is profitable.

Every symmetric profile has a profitable deviation, so none is a Nash equilibrium. (Intuition: below 5 the capacity-like supply limit leaves residual demand worth serving at the top price; above 5 a firm wants its whole supply p rather than half the market, 5.)

3. *Make a formal argument that asymmetric prices cannot be a pure strategy Nash equilibrium.*

Suppose $p_j < p_i$. The low-priced firm j earns $p_j^2/2$ by (A). Any $p_j' \in (p_j, p_i)$ still undercuts i, so j still sells its full supply, and it earns $p_j'^2/2 > p_j^2/2$. So j has a profitable deviation and (p_i, p_j) is not a Nash equilibrium. Combined with 2, there is no pure strategy Nash equilibrium.

4. *Use your result from 1 to find the bound of best response prices.*

From 1, every best (supremum) response to a pure price is either 10 or just below some $p_j > 5\sqrt{2}$. So best-response prices lie in

$$[\,5\sqrt{2},\ 10\,] \approx [7.07,\ 10].$$

The upper bound $\bar{p} = 10$ is the price a firm uses when it accepts being the high-priced firm: once it is the high firm, it should charge the most it can for the residual demand. Prices below $5\sqrt{2}$ are never a best response to a pure price, since against $p_j < 5\sqrt{2}$ the firm would rather jump to 10.

5. *Solve for the mixed strategy equilibrium using pricing bounds using the answer to 1.*

Look for a symmetric equilibrium in which each firm draws its price from a CDF F with support $[\underline{p}, 10]$.

No atoms. An atom at a price $p > 5$ cannot occur: the rival would gain by moving that mass to just below p, because tying earns $5p - 12.5 < p^2/2$. In particular there is no atom at 10, and none at $\underline{p}$.

Upper bound $\bar{p} = 10$. By (B), whenever a firm is the high-priced firm it is best off at 10, so the top of the support is 10.

Lower bound. At $\underline{p}$ a firm is the low-priced firm with probability 1, so its equilibrium payoff is $k = \underline{p}^2/2$. At 10 it is the high-priced firm with probability 1, so $k = E\big[(100 - X^2)/2\big]$ where $X \sim F$ is the rival's price. Indifference between the two ends gives

$$\frac{\underline{p}^2}{2} = 50 - \frac{E[X^2]}{2} \quad \Longleftrightarrow \quad E[X^2] = 100 - \underline{p}^2.$$

Note: $\underline{p}$ cannot equal the pure-strategy bound $5\sqrt{2}$ from 4. If it did, $E[X^2] = 50 = \underline{p}^2$, which would put all of F's mass at $\underline{p}$, and that contradicts no atoms. Against a *mixed* rival, prices somewhat below $5\sqrt{2}$ become worthwhile. So the bound is pinned down together with F (part 6): $\underline{p} \approx 6.60$.

6. *Solve for the Nash Equilibrium Mixed strategies.*

For $p \in [\underline{p}, 10]$ and a rival price $x < p$, the high firm sells $\min\{p, 10 - x\} = 10 - x$ (since $10 - x \le 10 - \underline{p} < \underline{p} \le p$). Expected profit from p is

$$\pi(p) = \int_{\underline{p}}^{p} \Big[p(10 - x) - \tfrac{1}{2}(10 - x)^2\Big] dF(x) + \big(1 - F(p)\big)\frac{p^2}{2} = k \quad \text{for all } p \in [\underline{p}, 10].$$

Differentiate once. The terms with $f(p)$ combine to $\big[p(10 - p) - \frac{1}{2}(10 - p)^2 - \frac{1}{2}p^2\big] f(p) = -2(p - 5)^2 f(p)$, so

$$\int_{\underline{p}}^{p} (10 - x)\, dF(x) + p\big(1 - F(p)\big) - 2(p - 5)^2 f(p) = 0.$$

Differentiate again and write $S = 1 - F$, $t = p - 5$:

$$2t^2 S'' + 6tS' + S = 0.$$

This is an Euler equation. Trying $S = t^r$ gives $2r^2 + 4r + 1 = 0$, so $r = -1 \pm \frac{1}{\sqrt{2}}$. Imposing $S = 0$ at $p = 10$ ($t = 5$) gives

$$F(p) = 1 - C\Big[5^{\sqrt{2}}(p - 5)^{-1 - \frac{1}{\sqrt{2}}} - (p - 5)^{-1 + \frac{1}{\sqrt{2}}}\Big], \qquad \underline{p} \le p \le 10.$$

The remaining two conditions are $F(\underline{p}) = 0$ and the first-derivative equation at $p = \underline{p}$, i.e. $\underline{p} = 2(\underline{p} - 5)^2 f(\underline{p})$. Solving them numerically:

$$\underline{p} \approx 6.602, \qquad C \approx 0.2869.$$

p	6.60	7	7.5	8	9	10
$F(p)$	0	0.378	0.635	0.780	0.929	1

The density is positive on the whole support, the mean price is about 7.47, and $E[X^2] \approx 56.41 = 100 - \underline{p}^2$, which checks the condition from 5. I also checked numerically that $\pi(p)$ is constant ($\approx 21.\overline{79}$) at every price in $[\underline{p}, 10]$. Below $\underline{p}$, $\pi(p) = p^2/2 < k$, so no price outside the support pays more.

The Nash equilibrium is: each firm independently draws its price from F on $[6.60, 10]$.

7. *Calculate the Nash equilibrium expected profits.*

$$E[\pi_i] = k = \frac{\underline{p}^2}{2} \approx \frac{6.602^2}{2} \approx 21.8 \text{ per firm.}$$

For comparison: two price-takers at $p = 5$ would each sell 5 and earn 12.5. Joint profit maximization (both at 10, each selling 5) would give each 37.5. Mixed pricing lands between the two.